Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program: BSCpE**
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed: 10/22/26**
Section: CPE11S4	**Date Submitted: 10/22/26**
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1. Objective/s:

This activity aims to navigate and manage files and directories using command line interface.

2. Intended Learning Outcome/s:

The students should be able to:

2.1 Demonstrate how to navigate home and system directories
2.2 Demonstrate how to list files and directories.
2.3 Demonstrate how to use globbing to manipulate files and directories.
2.4 Demonstrate how to create, move and delete files and directories.

3. Discussion:

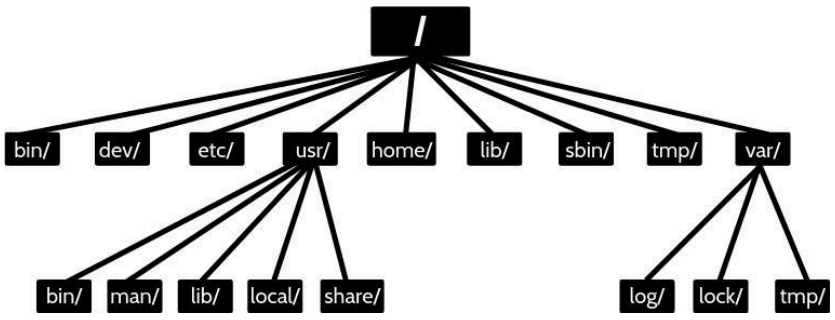
Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.

Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer.*** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory

On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory there is a directory for each user on the system. When a user opens a shell, they should

automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always starts at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

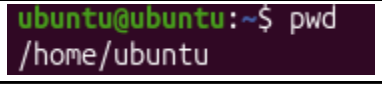
The single period . character always represents the current directory.
Two period .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

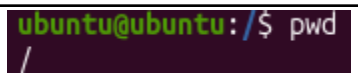
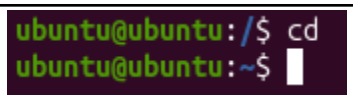
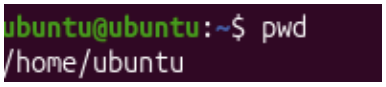
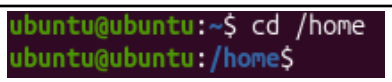
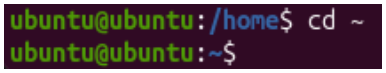
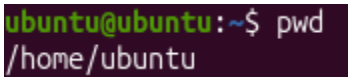
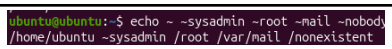
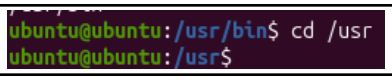
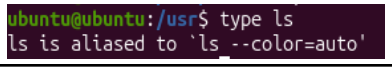
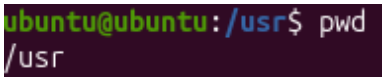
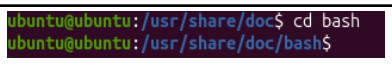
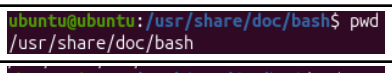
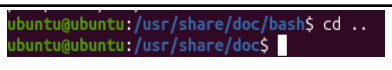
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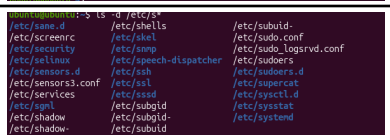
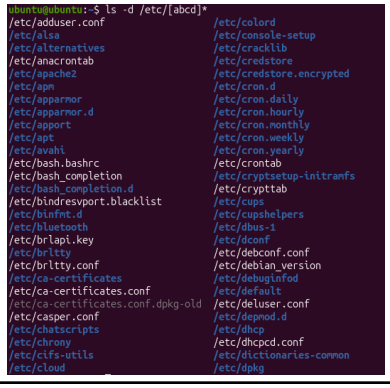
Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

5. Procedure:

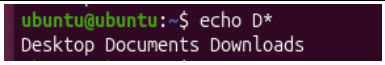
1. Login using your username and password.
2. Use terminal emulator application (if you are using desktop version)
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

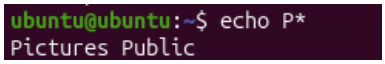
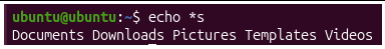
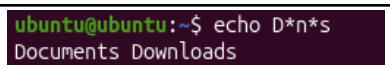
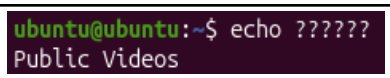
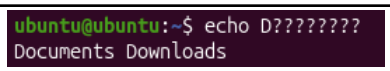
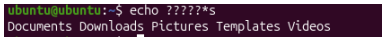
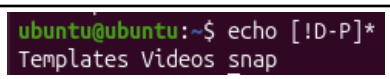
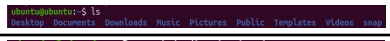
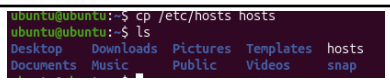
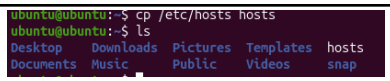
Command	Screenshot	Explanation
1. pwd		it shows the current directory path

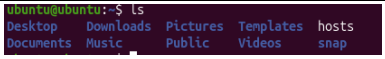
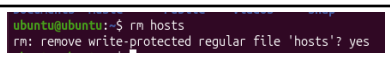
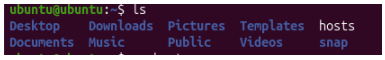
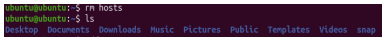
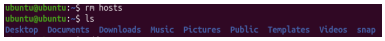
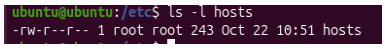
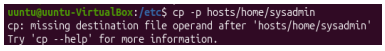
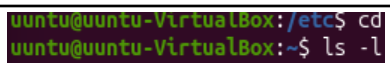
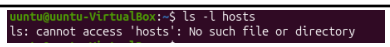
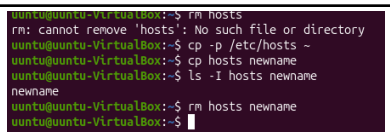
2. cd /		Takes you straight to the root directory
3. pwd		It prints the current directory
4. cd		moves you between folders in your terminal
5. pwd		it displays the current path
6. cd /home		changes your current directory to the home directory
7. pwd		It moves you to the home directory
8. cd ~		It shows you the current folder
9. pwd		it brings you back to the home directory
10. echo ~ ~sysadmin ~root ~mail ~nobody where sysadmin is your username		prints the home directories of the specified user accounts to the standard output
11. cd ~root		Denied because the account tried to access a protected directory
12. cd /usr/bin		It is the directory inside the file system
13. pwd		Prints out your current directory /usr/bin
14. cd /usr		Changes your directory to /usr
15. type ls		it changes the color to auto
16. pwd		Prints the current directory
17. cd /usr/share/doc		Brings you to the directory of the documentations files of the user
18. pwd		Prints out the current directory
19. cd bash		brings the user to the directory of the official documentation for the bash
20. pwd		Prints the current directory
21. cd ..		It goes back to the directory before it

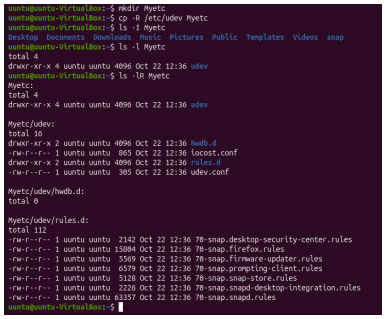
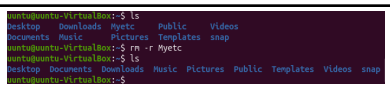
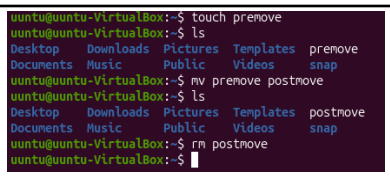
22. pwd		Prints the current directory
23. cd ../dict		Go to dict folder in parent directory
24. pwd		Shows current directory location
25. cd		Changes the directory and waits for the user's input
26. ls		it list down all the directory
27. ls -a		List all files and also hidden files starting with '.'
28. ls -l /etc/hosts		it is built in shell command on external file
29. ls -R /etc/udev		recursively lists all files, scripts, and rule directories inside the Linux
30. ls -d /etc/s*		lists only the names of directories and files inside the /etc folder that begin with the letter "s,
31. ls -d /etc/????		lists all four-character files or subdirectories located directly inside the /etc configuration directory,
32. ls –d /etc/[abcd]*		lists specific directories and files inside the /etc/ folder that start with the letters a, b, c, or d, without showing the contents of those subdirectories.

4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

Command	Screenshot	Explanation
1. echo *		Displays all files and directories in the current directory.
2. echo D*		Displays all files/directories starting with capital letter D.

3. echo P*		Displays all files/directories starting with capital letter P.
4. echo *s		Displays all files/directories starting with capital letter s.
5. echo D*n*s		Displays files starting with D, containing n in the middle, and ending with s.
6. echo ??????		The ? matches exactly one character . Six ? means files with exactly 6 letter names like public and videos.
7. echo D????????		Matches 10 letter names starting with D. D+9 ? = 10 characters.
8. echo ?????*s		Matches files with at least 5 characters and ending with s.
9. echo [DP]*		Bracket expansion. Matches files starting with either D or P only.
10. echo [!DP]*		Range expansion. Matches files starting with any letter from D to P. Shows Desktop,documents,downloads, music, pictures,public.
11. echo [D-P]*		uses a pattern-matching feature called globbing to search for and list files or folders in your current directory that start with specific capital letters.
12. echo [!D-P]*		uses a pattern-matching feature of the shell called filename expansion
13. ls		List down all the files
14. cp /etc/hosts hosts		copies the system's local hosts configuration file and saves a duplicate named host in your current working
15. ls		List down all the files
16. rm hosts		deletes the hosts file from your current directory.
17. ls		List down all the files
18. cp –v /etc/hosts hosts		copies the system's hosts file to your current directory while

		displaying exactly what it is doing.
19. ls		List down all the files
20. rm hosts		permanently delete a file named "hosts" from your current working directory.
21. ls		List down all the files
22. cp –v /etc/hosts .		copies the system's hosts file to your current working directory while displaying what it is doing in the terminal.
23. ls		List down all the files
24. rm hosts		permanently delete the hosts file from your computer.
25. ls		List down all the files
26. cd /etc		command changes your current working directory
27. ls -l hosts		is used in Linux and Unix-like operating systems to display detailed information about a specific file named
28. cp –p hosts/home/sysadmin		copies a file while preserving its original attributes, but it contains a syntax error that will prevent it from running correctly.
29. cd		Return to home directory
30. ls –l hosts		Verify file was copied with same permissions
31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname 34. ls –l hosts newname 35. rm hosts newname		31.in your current working directory. If the file does not exist, it will return an error. 32.it preserves the original file's attributes, such as its original modification time, access time, and file permissions. 33 file in the current directory and names the copy 34.for both the hosts and newname files. It shows their permissions, owners, file sizes, and modification dates so you can verify they are identical.

		35.files from the current directory, cleaning up the files created in the previous steps.
36. mkdir Myetc 37. cp –R /etc/udev Myetc 38. ls –l Myetc 39. ls –lR Myetc		36. Creates a new directory named Myetc in your current working directory. 37. which contains system device management rules 38. Lists the immediate contents of the Myetc directory This view shows detailed information for the copied udev folder, including file permissions, owner, group, file size, and the date it was modified. 39. forces the command to dive into every subdirectory inside
40. ls 41. rm -r Myetc 42.ls		40. List file in current directory 41. Delete directory Myetc recursively 42.
43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		43. Create empty file named premove 44. Verify file creation 45. Rename/move file from premove to postmove 46. Verify rename 47. Delete the postmove file

6. Supplementary Activity:

```
uuntu@uuntu-VirtualBox:~$ mkdir -p ~/Documents/CPE231
uuntu@uuntu-VirtualBox:~$ cd ~/Documents/CPE231
uuntu@uuntu-VirtualBox:~/Documents/CPE231$ mkdir Pascual Matt
uuntu@uuntu-VirtualBox:~/Documents/CPE231$ cd Pascual
uuntu@uuntu-VirtualBox:~/Documents/CPE231/Pascual$ touch A.txt B.txt
uuntu@uuntu-VirtualBox:~/Documents/CPE231/Pascual$ cd ~/Documents
uuntu@uuntu-VirtualBox:~/Documents$ mkdir CPE231_backup
uuntu@uuntu-VirtualBox:~/Documents$ cp -r CPE231/* CPE231_backup/
uuntu@uuntu-VirtualBox:~/Documents$ rm -rf ~/Documents/CPE231
uuntu@uuntu-VirtualBox:~/Documents$
```

7. Conclusion:
In this activity, I learned how to navigate and manage files and directories in Linux using terminal commands such as pwd, cd, ls, mkdir, cp, mv, and rm. I also learned how to use globbing to find files and directories. Overall, this activity helped me improve my understanding of basic Linux commands and file management.
8. Assessment (Rubric for Laboratory Performance):